Ezra Apple

Software Engineer | AI Products & Agent Systems

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Education

University of California, Berkeley May 2025
B.S. in Electrical Engineering and Computer Science

Experience

Replo (YC S21) | *Software Engineer, AI Agent Platform* Oct 2025 – Oct 2026

- Owned **reusable-content workflows** for Replo's AI e-commerce platform, from service APIs and authentication through editor integration and live previews. Built the Component Library and template installation to let merchants reuse React storefront components across pages, contributing to the team's beta-to-GA launch.
- Designed and ran **agent evaluations** using customer-derived and synthetic editing tasks, comparing context and instruction changes on edit-target correctness, cost, and latency.
- Redesigned **persistent agent memory** with scoped user/project context, compact working indexes, and automatically curated topic pages retrieved on demand; shipped a memory browser for inspecting saved context.
- Hardened **cloud-sandbox editing and live previews** against stale builds and styles, adding runtime checks, bounded automatic recovery, actionable diagnostics, and preview restart controls.
- Led consolidation of Replo's **agent skills and development workflows**, organizing engineering, design, and support guidance; reduced the agent-facing catalog from 75 to 42 skills and added authoring standards, CI checks, and routing tests across Claude Code, Codex, Cursor, and opencode.

Decyphr AI | *Co-Founder & CTO* Jan 2025 – Aug 2025

- Co-founded AI video translation startup, accepted into **UC Berkeley SkyDeck Pad-13** incubator program.
- Architected an **asynchronous media-processing pipeline on AWS** (S3, Lambda, Step Functions, SES) with fault-tolerant retries for long-running translation jobs and user-visible processing and failure states.
- Owned an **end-to-end video-translation MVP** combining **ElevenLabs dubbing** with lip-sync, spanning authentication, media upload, processing orchestration, and launch site.
- Ran a **closed beta with 10+ content creators**, using their feedback to refine the product and go-to-market.

Lawrence Berkeley National Laboratory | *Machine Learning Researcher* Oct 2023 – Jun 2025

- Applied **random forests and Bayesian optimization** to ion-source diagnostic data, modeling accelerator performance and exploring configurations to maximize ion output.
- Presented findings to senior researchers supporting **heavy-element synthesis**, including the element 120 program.

Projects

ShoutOut | *Local-First macOS Dictation App*

- Built and shipped a local-first macOS dictation app in **Swift** that inserts transcribed speech into other apps, with signed, notarized releases and automatic updates.
- Fine-tuned **BERT Mini** as a KEEP/DELETE token classifier for dictation cleanup, then exported it to **Core ML** and integrated on-device inference into the app's transcription pipeline.
- Streamed microphone PCM to a local **Parakeet/Metal** transcription helper with session-based capture, final decoding on completion, and a local full-clip fallback.

Skills

Languages: TypeScript, Python, Swift, SQL
AI & Agents: Agent workflows, agent evaluation, persistent memory, internal agent skills, MCP integrations
ML & Data: PyTorch, Hugging Face Transformers, scikit-learn, NumPy, pandas
Backend & Infrastructure: Node.js, tRPC, PostgreSQL, Cloudflare Workers, AWS, Docker, GitHub Actions
Frontend & Desktop: React, Next.js, Tailwind CSS, SwiftUI, macOS